

Wildfire smoke increases labour hours among urban workers

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Abstract

Wildfire-smoke-driven increases in fine particulate matter raise hours worked and employment in Canada's nine largest metropolitan areas. This response is consistent with productivity impairment requiring longer hours to meet output targets and with degraded outdoor conditions reducing the value of non-work time. Both mechanisms imply welfare losses that standard labour-market statistics record as gains, suggesting output-based damage assessments understate the costs of smoke in urban economies.

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Wildfire smoke reduces employment and earnings, particularly among outdoor labourers and low-income workers, where direct physical exposure and employer risk management lead to reduced hours (1; 2). Rising wildfire intensity across developed countries, including eightfold increases in area burned across much of North America since 1985 (3), has begun to erode decades of air-quality progress (4) and generates transboundary smoke events affecting populations thousands of kilometres from active fires (5; 6). Wildfire-specific airborne fine-particulate matter (PM_{2.5}) carries greater respiratory risk than equivalent urban-source concentrations (7), and global evidence attributes meaningful excess mortality to acute exposure (8). While existing studies have focused primarily on outdoor, low-income, and aggregate labour-market effects, how wildfire smoke affects labour supply differently in the predominantly indoor, salaried workforces that dominate wealthy urban economies remains largely unexamined. In developed countries, metropolitan areas typically account for the majority of economic output (three-quarters of GDP in Canada, for example), with workforces concentrated in service and knowledge sectors, and where climate-controlled indoor work is the norm. The labour-market consequences of wildfire smoke in these settings remain unknown.

We evaluate the impact of wildfire smoke exposure in Canada’s nine largest Census Metropolitan Areas using monthly *Canadian Labour Force Survey* (LFS) microdata (2015–2024, excluding 2020). The analysis instruments ambient PM_{2.5} with satellite-detected wildfire smoke plumes during the May–October fire season. The atmospheric transport of distant wildfire smoke provides plausibly exogenous variation in local air quality, and the relevance of the instrument is confirmed by a first-stage F-statistic well above conventional thresholds ($F > 100$), indicating that smoke-day counts are a strong predictor of ground-level PM_{2.5} concentrations. Contrary to findings from more rural, outdoor, and lower-income settings (e.g., 2), in the urban labour markets of our study, a 10 $\mu\text{g}/\text{m}^3$ increase in smoke-driven PM_{2.5} significantly *increases* hours worked (+0.031 log points, an approximate +3.1% increase, $p < 0.05$; Table 2).¹ Employment and labour-

¹Because the outcome is $\log(\text{hours} + 1)$, the percentage effect is calculated as $100 \times (e^{\hat{\theta}} - 1)$ rather than the linear approximation $100 \times \hat{\theta}$, which is only accurate for small coefficients. For $\hat{\theta} = 0.031$: $100 \times (e^{0.031} - 1) \approx 3.1\%$. The same formula is used for all other log-point figures reported in percentage terms throughout the paper.

force participation rise (+1.2 and +0.8 percentage points, respectively; all reported values are per 10 $\mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ increase). The newly employed, if entering at fewer-than-average hours, would attenuate rather than contribute to the average-hours increase. Formal absences decline slightly (-0.007 , $p < 0.05$). Earnings rise significantly as well (+0.018 log points, an approximate +1.8% increase, $p < 0.05$), which complicates a pure flat-wage interpretation: the combination of higher hours, higher employment, and higher earnings is consistent with either overtime-type compensation for additional hours or a compositional shift toward higher-paying employment during smoke events, and we cannot fully distinguish these channels with the current specification.²

Industry-level estimates show that the positive hours response is broadly shared across the urban economy (Table 3). Using manufacturing as the baseline, net point-estimate hours effects are positive for most industries (13 of 17), including retail, professional services, healthcare, construction, business support, and finance, though several of these net effects are modest in size. Agriculture, Forestry & Fishing and Utilities show small net declines (net -0.031 log points, an approximate -3.1% , and net -0.030 log points, an approximate -3.0% , respectively), consistent with employee- or employer-driven avoidance of physical labour and exposure-sensitive maintenance work in high-exposure environments. Public Administration’s net effect is essentially null (net -0.001 log points). Education shows a much larger net decline (net -0.565 log points, an approximate -43.2%), consistent with school closures as many urban school buildings without central air conditioning cannot maintain safe indoor air quality during high-smoke events, and Canadian school boards issued documented closures and early dismissals during major fire seasons across the study period (9). Beyond these cases, manufacturing’s own effect is not significantly different in magnitude from most other industries’ responses: only five of the other 16 industries—Utilities, Finance, Education, Healthcare, and Public Administration—have interaction terms significantly different from zero, indicating the positive hours effect is broadly shared across the urban economy rather than concentrated

²We re-estimate the industry specifications using instruments restricted to medium and heavy smoke plumes only, and then to heavy plumes exclusively. Effects are consistent under all three density thresholds; only the earnings coefficient attenuates toward zero under the stricter definitions, consistent with reduced first-stage variation rather than a qualitatively different effect (see Supplementary Methods)

narrowly in manufacturing. This is consistent with the paper’s central claim: hours rise on average across the urban economy, with Education standing out as the clearest and largest exception, driven by an institutional (school-closure) channel rather than the broader indoor-economy mechanism.

Prior studies interpreted weak or null labour-market effects among indoor and higher-wage workers as due to their insulation from environmental harm (2). Our results directly challenge this interpretation. In Canada’s largest cities, where indoor salaried employment dominates, work hours increase with smoke exposure. The workers are not insulated from impact. They are affected differently than outdoor workers, through channels that conventional labour-market metrics record as gains rather than losses.

To see why more hours do not mean higher well-being, consider stories of two workers on a smoky Friday afternoon. One worker has had headaches and been struggling to concentrate as smoke has drifted through the city for days. His team is slightly behind at the end of the week, and he stays late to catch up. He spends extra time at work, but accomplishes nothing more than he would have without the smoke-driven productivity decline. A second worker planned to leave work early to meet friends on a patio. The smoke leads to canceled plans, and she stays at her desk. While she completes more work that afternoon, it is second best for her, as without the smoke, she would have chosen the patio over the extra work. The smoke has not created value; it has devalued her leisure time, leading to a net welfare loss despite the extra work she completes. In both cases, hours go up. In neither case is the worker better off.

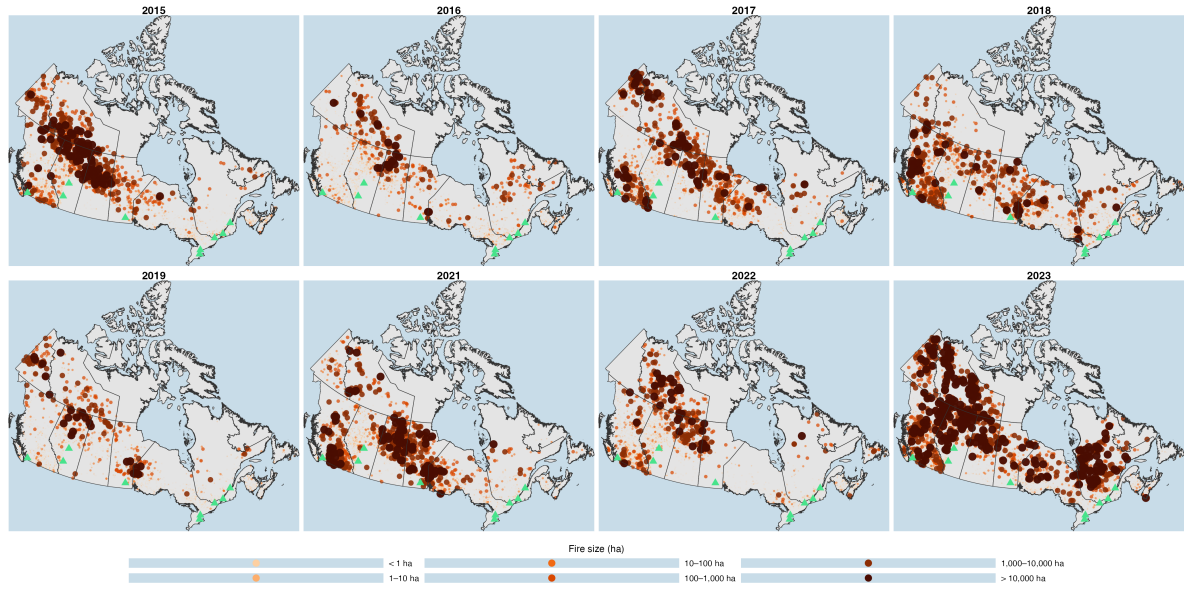
These stories highlight two mechanisms that are observationally equivalent in our data and likely operating together. First, $PM_{2.5}$ impairs cognitive and physical output even in enclosed workplaces (10; 11; 12); workers who accomplish less per hour may stay longer to meet unchanged output expectations. Second, smoke degrades outdoor recreation, leisure, and home activity for those without adequate filtration, lowering the value of non-work time and the opportunity cost of working more, and drawing some workers into climate-controlled offices to reduce exposure (13). Both predict more hours without a welfare gain: under the first, added hours produce no added output; under

the second, output may rise only because leisure is degraded. Hours, employment, and earnings record no loss.

This is a welfare-output disconnect that standard damage assessments are not built to detect. Prior work treats market labour statistics as a lower bound on smoke’s welfare costs, since productivity losses and avoidance behavior need not surface in earnings (2). We find that for urban workers, the bound not only understates the cost but also has the wrong sign. The harm registers as a gain. And it registers unevenly: those with high-quality home filtration have different options for minimizing exposure outside of work than others, a disparity that hours and earnings data cannot capture.

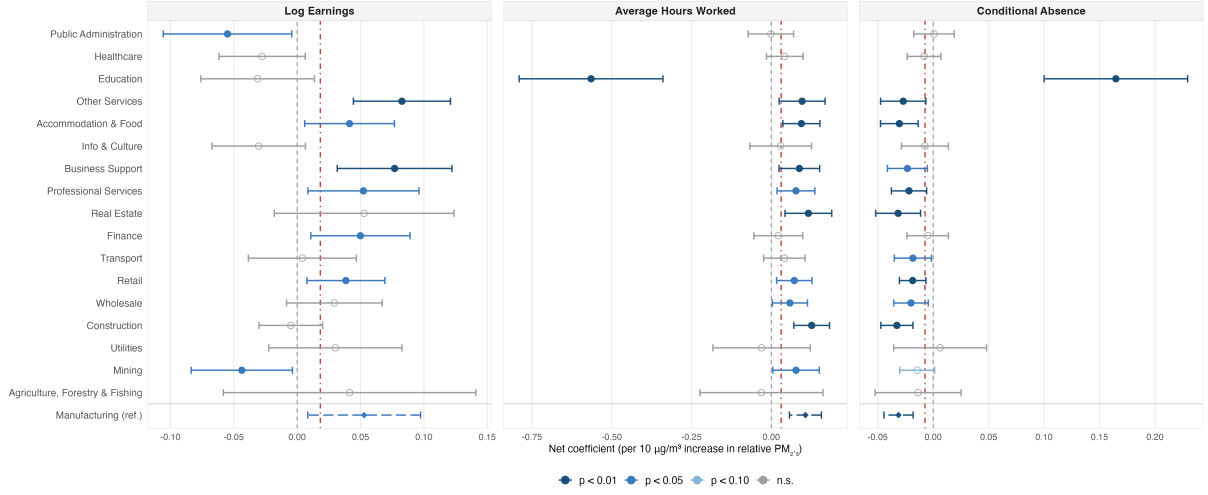
These mechanisms are likely features of indoor, service-sector work in high-income economies, not artifacts of Canadian institutions. As wildfire activity intensifies (3) and smoke-attributable mortality climbs under continued warming (14), integrated assessment models that price environmental costs through earnings or GDP (15) may misread the indoor economy entirely: workers put in more time, earnings barely move, and output may or may not rise. A credible damage metric must capture what these models omit: the value of non-work time that smoke destroys, and the output gap left when longer hours merely offset lost productivity. As smoke becomes a fixture of the climate across the global north, getting these metrics right may help societies assess the true (and heterogeneous) cost of a warming world.

Figure 1: Recorded wildfires across Canada during the May–October fire season, 2015–2023



Each panel plots fire ignition locations from the Canadian National Fire Database (Natural Resources Canada) for the indicated year. Dot size and colour indicate fire size category: small fires (< 1 ha, orange) through large fires (> 10,000 ha, dark brown). Green triangles mark Canada's nine largest Census Metropolitan Areas, which form the study sample. The figure illustrates the geographic breadth and interannual variability of wildfire activity, and the spatial separation between fire sources and the urban study sites—a key feature supporting the identifying assumption that smoke transport reflects distant fire activity rather than local emissions.

Figure 2: Net PM_{2.5} Effects by Industry



Point estimates and 95% confidence intervals from two-stage least squares regressions on instrumented PM_{2.5}, interacted with industry indicators, showing each industry's net effect (the baseline PM_{2.5} coefficient plus that industry's interaction term) rather than the interaction term alone. Manufacturing (diamond marker) is the omitted reference category, so its point reflects the baseline PM_{2.5} coefficient alone; confidence intervals for all other industries account for the covariance between the baseline and interaction coefficients. Outcomes are log monthly earnings (left), average hours worked in the main job, reported as $\log(\text{hours} + 1)$ (centre), and a conditional absence indicator (right). Coefficients are scaled per 10 $\mu\text{g m}^{-3}$ increase in relative PM_{2.5}. Colour and fill indicate statistical significance of the net effect: solid dark blue $p < 0.01$, solid blue $p < 0.05$, light blue $p < 0.10$, grey not significant. The dotted red line marks the pooled population-average baseline estimate (Table 2). Sample: Canada's nine largest CMAs, May–October 2015–2024 excluding 2020. Standard errors clustered at the CMA $\times$ year level.

Online Methods Section

We study short-run labour-market responses to wildfire-driven air pollution in Canada’s nine largest Census Metropolitan Areas over 2015–2024, restricting the sample to the May–October fire season when wildfire smoke is the dominant source of transient $\text{PM}_{2.5}$ variation (16). The year 2020 is excluded to avoid confounding from COVID-19 labour-market disruptions; pre-COVID and post-COVID subsamples yield consistent results.

Individual labour-market outcomes are drawn from monthly microdata in Statistics Canada’s Labour Force Survey (17). Primary outcomes include employment status, labour-force participation, log monthly earnings, log hours worked in the main job, and a conditional absence indicator (equal to one for employed individuals who were absent from work during the reference week). Hours worked and hourly earnings are defined by Statistics Canada for employees only; self-employed workers and unpaid family workers are excluded from these two outcomes, though not from employment, labour-force participation, or the absence indicator, which are defined for all labour-force participants. Job characteristics—union coverage, public-sector employment, full-time status, and firm size—are defined from standard LFS identifiers. All regressions are weighted by LFS final survey weights.

Ambient $\text{PM}_{2.5}$ concentrations ($\mu\text{g}/\text{m}^3$) are obtained from Environment and Climate Change Canada’s National Air Pollution Surveillance network (18). For each CMA-month, we compute mean daily $\text{PM}_{2.5}$ over a 30-day window ending on the 15th of the survey month, aligning the exposure period with the LFS reference week (the week containing the 15th of each month). We then demean this measure within each CMA by subtracting the CMA-specific pooled mean across the full sample period, yielding relative $\text{PM}_{2.5}$ ($\widetilde{PM}_{ct}$). This isolates transient pollution shocks from persistent local air-quality differences; remaining seasonal variation is absorbed by $\text{CMA} \times \text{calendar-month}$ fixed effects in the regression.

We employ an instrumental-variables design following Borgschulte et al. (author?) (2), using satellite-detected wildfire smoke plumes from NOAA’s Hazard Mapping System (19) as an instrument for ambient $\text{PM}_{2.5}$. A smoke day is defined as any day on which a

detected plume intersects the boundary of a CMA. The smoke-day count is measured over the same 30-day window as $PM_{2.5}$ and demeaned within each CMA to form relative smoke days ($\widetilde{Smoke}_{ct}$). The identifying assumption is that, conditional on CMA $\times$ calendar-month and year fixed effects plus meteorological controls (temperature and precipitation), atmospheric smoke transport affects labour outcomes only through its impact on local $PM_{2.5}$. This assumption is most credible in large urban areas, where smoke exposure reflects distant fire activity rather than local emissions. The first stage is estimated as:

$$\widetilde{PM}_{ct} = \alpha + \beta \widetilde{Smoke}_{ct} + X_{ct}\gamma + \mu_{cm} + \tau_t + \varepsilon_{ct} \quad (1)$$

where $\widetilde{Smoke}_{ct}$ is the CMA-demeaned smoke-day count over the 30-day reference window, X_{ct} are meteorological controls (mean temperature and total precipitation), μ_{cm} are CMA $\times$ calendar-month fixed effects, and τ_t are year fixed effects. The second stage relates individual labour outcomes to instrumented $PM_{2.5}$:

$$Y_{ict} = \delta + \theta \widetilde{PM}_{ct} + Z_{ict}\lambda + \mu_{cm} + \tau_t + \eta_{ict} \quad (2)$$

All coefficients are scaled to reflect a 10 $\mu g/m^3$ increase in relative $PM_{2.5}$. Standard errors are clustered at the CMA $\times$ year level. Heterogeneous effects are estimated by interacting instrumented $PM_{2.5}$ with industry indicators. Robustness analyses, including alternative fixed-effect structures, exposure definitions, and demographic and job-characteristic interactions, are reported in the Supplementary Information.

We assess robustness to the definition of a smoke exposure day using density classifications from the NOAA HMS product, which codes each detected plume as Light, Medium, or Heavy based on satellite-assessed aerosol optical depth. Our baseline instrument counts all plume-intersecting days regardless of density. We construct two alternative instruments: one counting only Medium and Heavy days, and one counting only Heavy days. Both alternatives use the same 30-day window and CMA-demeaning procedure as the baseline. The baseline industry-specification hours effect of +0.107 log points, an approximate +11.3% ($p < 0.01$), remains positive and statistically significant

when restricted to Medium and Heavy days (+0.086 log points, an approximate +9.0%, $p < 0.01$) and to Heavy days only (+0.067 log points, an approximate +6.9%, $p < 0.01$). The conditional absence effect is likewise unchanged across all three thresholds (−0.031, −0.026, and −0.021 log points, respectively, all $p < 0.01$). The earnings coefficient, significant at the 5% level in the baseline industry specification (+0.053 log points, an approximate +5.4%), attenuates and loses significance under the two stricter definitions (+0.018 and +0.004, both statistically insignificant)—a pattern consistent with noisier identification as the instrument discards lighter plume days rather than a genuine difference in the earnings channel.

The main analysis excludes 2020 to avoid confounding from the acute COVID-19 labour-market disruptions. Pandemic-related layoffs, sector-specific shutdowns, and the rapid expansion of remote work might contaminate inference about the pollution-hours relationship. To verify that the remaining results are not specific to either the pre-pandemic or post-pandemic period, we re-estimate the primary specifications separately for the pre-COVID subsample (2015–2019) and the post-COVID subsample (2021–2024).

Table 1 reports the baseline PM_{2.5} coefficients for hours worked (in logs), conditional absence, and log earnings across the two sub-periods. The positive hours response and the negative absence effect are statistically significant in both sub-periods, with no sign reversal. The hours coefficient is larger in the pre-COVID period (+0.084 log points, an approximate +8.8%, $p < 0.01$) than in the post-COVID period (+0.030 log points, an approximate +3.1%, $p < 0.05$), a difference consistent with the post-period sample being dominated by years of unusually intense wildfire activity—2021, 2022, and 2023—in which urban workers may have partially adapted avoidance strategies not present in earlier, lower-baseline seasons. Earnings remain statistically indistinguishable from zero in both sub-periods individually; since the pooled estimate (Table 2) is significant, this most likely reflects reduced power from halving the sample rather than a genuine difference in the earnings channel across periods. These findings confirm that the hours and absence results are a persistent feature of the data across distinct macroeconomic regimes rather than an artifact of conditions in either half of the sample.

Table 1: Before and after COVID robustness.

	Average Hours		Absent		Earnings	
	Pre	Post	Pre	Post	Pre	Post
PM _{2.5}	0.084*** (0.031)	0.030** (0.013)	-0.024*** (0.008)	-0.010*** (0.004)	-0.003 (0.007)	-0.003 (0.006)
Observations	469,515	440,310	469,515	440,310	469,515	440,310

Notes: IV estimates of smoke-driven PM_{2.5} on labour-market outcomes, estimated separately for pre-COVID (2015–2019) and post-COVID (2021–2024) subsamples, both drawn from the 2020-excluded sample. Coefficients scaled per 10 $\mu\text{g}/\text{m}^3$. Average Hours is $\log(\text{hours} + 1)$; see Table 2 notes. Specifications include job-characteristic controls (union coverage, public-sector status, full-time status, and firm size) in addition to CMA $\times$ month and year fixed effects and meteorological controls. Standard errors clustered at the CMA $\times$ year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: IV estimates of smoke-driven PM_{2.5} on labour-market outcomes

	(1) Employment	(2) LFP	(3) Earnings	(4) Average Hours	(5) Absent
PM _{2.5}	0.012*** (0.004)	0.008* (0.004)	0.018** (0.009)	0.031** (0.014)	-0.007** (0.003)
Observations	1,700,116	1,700,116	909,825	1,064,322	1,064,322

Notes: Two-stage least squares estimates using wildfire smoke days as instrument for PM_{2.5}. Coefficients scaled per 10 $\mu\text{g}/\text{m}^3$ increase in relative PM_{2.5}. Column (4) reports the outcome as $\log(\text{hours} + 1)$ rather than levels, since average hours worked has a meaningful mass point at zero (employed individuals absent the full reference week); the coefficient is therefore approximately the proportional effect on hours. Sample: nine largest Canadian CMAs, May–October 2015–2024 excluding 2020. All specifications include CMA $\times$ month and year fixed effects and meteorological controls. Standard errors clustered at CMA $\times$ year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: IV Estimates with Industry Controls

	(1) Earnings	(2) Average Hours	(3) Absent
PM _{2.5}	0.053** (0.023)	0.107*** (0.026)	-0.031*** (0.007)
PM _{2.5} × Agriculture, Forestry & Fishing	-0.012 (0.056)	-0.138 (0.104)	0.018 (0.020)
PM _{2.5} × Mining	-0.097*** (0.028)	-0.029 (0.039)	0.017* (0.009)
PM _{2.5} × Utilities	-0.023 (0.031)	-0.137* (0.082)	0.037 (0.024)
PM _{2.5} × Construction	-0.058*** (0.021)	0.020 (0.032)	-0.001 (0.008)
PM _{2.5} × Wholesale	-0.024 (0.030)	-0.049 (0.029)	0.011 (0.008)
PM _{2.5} × Retail	-0.014 (0.024)	-0.035 (0.030)	0.013 (0.008)
PM _{2.5} × Transportation	-0.049** (0.021)	-0.066 (0.040)	0.013 (0.011)
PM _{2.5} × Finance	-0.003 (0.027)	-0.085* (0.045)	0.026** (0.012)
PM _{2.5} × Real estate	-0.000 (0.048)	0.009 (0.040)	-0.000 (0.012)
PM _{2.5} × Professional services	-0.001 (0.037)	-0.030 (0.036)	0.009 (0.009)
PM _{2.5} × Business support	0.024 (0.020)	-0.019 (0.036)	0.008 (0.009)
PM _{2.5} × Education	-0.084*** (0.025)	-0.672*** (0.124)	0.196*** (0.036)
PM _{2.5} × Healthcare	-0.081*** (0.029)	-0.065* (0.036)	0.023** (0.010)
PM _{2.5} × Information & culture	-0.083*** (0.028)	-0.078 (0.048)	0.024** (0.011)
PM _{2.5} × Accommodation & food	-0.012 (0.020)	-0.012 (0.032)	0.001 (0.008)
PM _{2.5} × Other services	0.030 (0.022)	-0.010 (0.040)	0.004 (0.010)
PM _{2.5} × Public administration	-0.108*** (0.034)	-0.108** (0.042)	0.032*** (0.012)
Industry controls	Yes	Yes	Yes
Observations	909,825	1,064,322	1,064,322

Notes: IV estimates using wildfire smoke as instrument for PM_{2.5}. Sample excludes 2020. Omitted category is manufacturing. Agriculture, Forestry, and Fishing are combined into one category; Mining is estimated separately. Hours and Earnings are reported as log(hours + 1) and log(earnings), respectively; see Table 2 notes. All regressions include weather controls and CMA×month and year fixed effects. Standard errors clustered at the CMA-by-year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Key contrasts: Point-estimate net effects (baseline + interaction, in log points) are positive for 13 of 17 industries, though several are modest in size (e.g., Finance, Information & Culture, and Transportation are all under 0.05 log points). Four sectors show negative net effects: Agriculture, Forestry & Fishing (net -0.031 log points, an approximate -3.1%), Utilities (net -0.030 log points, an approximate -3.0%), Public Administration (net -0.001 log points, essentially null), and Education (net -0.565 log points, an approximate -43.2%, by far the largest deviation from baseline), consistent with outdoor/physical exposure, exposure-sensitive maintenance work, and smoke-related school closures respectively. Beyond these cases, manufacturing's own response is not significantly different in magnitude from most other industries: only Utilities, Finance, Education, Healthcare, and Public Administration have interaction terms significantly different from zero (5 of 17), indicating the aggregate positive effect is broadly shared across the urban economy rather than concentrated narrowly in manufacturing. Net-effect magnitudes are point estimates only; formal significance of each net effect requires a joint (linear-combination) test using the full covariance matrix, not reported here.

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